Trainings

Preparatory Steps

⚠ **WARNING** ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arching equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ **WARNING** ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

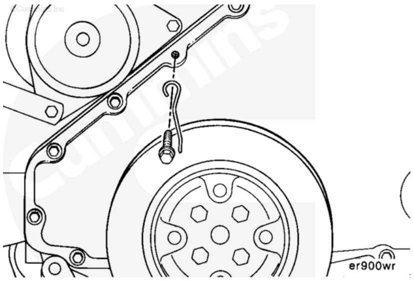
- Clean any debris from the fuel injection pump.
- Remove the fuel supply lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/41/41-006-024.html).
- Remove the injector supply lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html).
- Remove the control linkage. Refer to the OEM service manual.
- Remove the fuel shutoff solenoid. Refer to Procedure 005-043 (/qs3/pubsys2/xml/en/procedures/41/41-005-043.html).
- Remove the air fuel control air line.
- Remove the governor oil supply line.

Time

Marine Applications

⚠ CAUTION ⚠

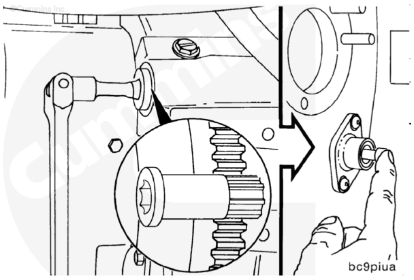
Do not use the timing pin to locate engine top dead center for marine applications. This can result in incorrect timing, poor engine performance, and engine smoke problems.



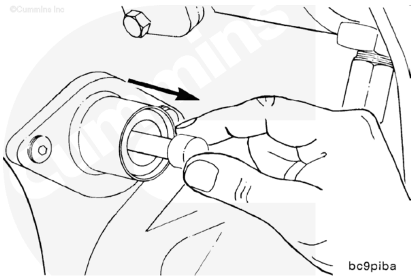
To locate actual top dead center, fabricate a timing mark pointer for the front of the engine.

Note : This can be done by forming a piece of 16 gauge wire that can be tightened under one of the gear cover capscrews. Sharpen the wire at the vibration damper end so that it comes to a point for better accuracy.

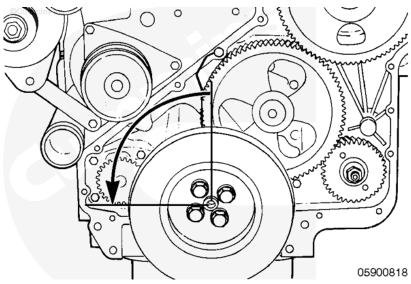
Locate top dead center for cylinder Number 1 by barring crankshaft slowly while pressing on the engine timing pin. Barring the engine is recommended from the flywheel on the rear of the engine using barring tool, Part Number 3824591.



To prevent damage to the engine or timing pin, you **must** disengage the timing pin after locating top dead center.

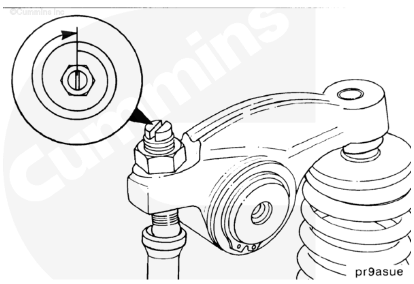


Rotate the crankshaft 90 degrees in the direction opposite normal rotation (**counterclockwise**).



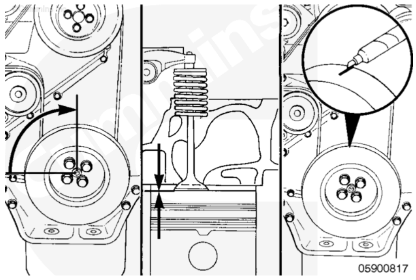
Tighten the adjusting screw for the intake valve on number 1 cylinder three complete turns of the screw.

Note : Leave the adjusting screw in this position until top dead center is established.



⚠ CAUTION ⚠

Use extreme care that the piston does not push against the valve with so much force that it bends the push rod or valve.

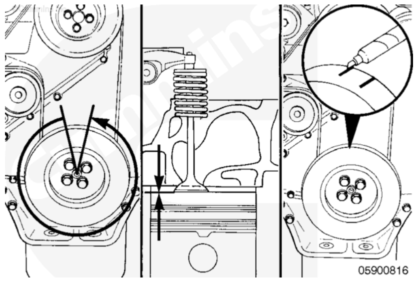


Rotate the crankshaft slowly in the direction of the engine rotation until the piston touches the intake valve.

Place a mark on the vibration damper at the tip of the pointer.

⚠ CAUTION ⚠

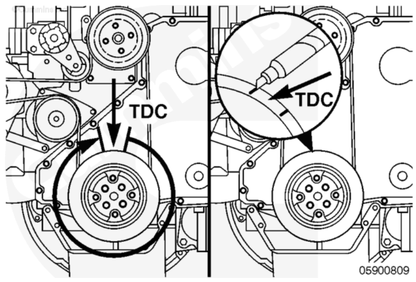
Make certain that the piston does not push against the valve with so much force that it bends the push rod or valve.



Rotate the crankshaft in the opposite direction until the piston touches the intake valve.

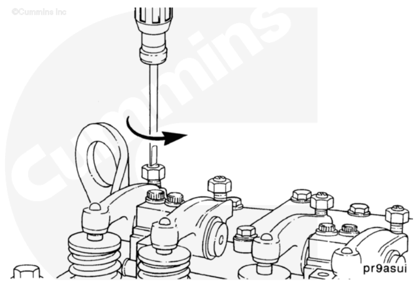
Place a mark on the vibration damper at the tip of the pointer.

Measure the distance and mark the damper at one-half the distance between the two marks. This mark is the top dead center mark.



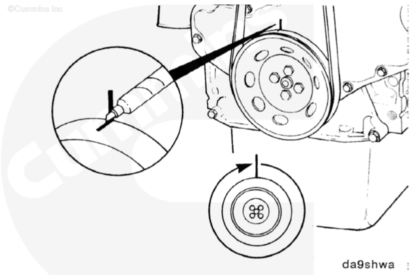
⚠ CAUTION ⚠

Failure to loosen the adjusting screw will result in bending the push rod and/or valve stem when the crankshaft is rotated.

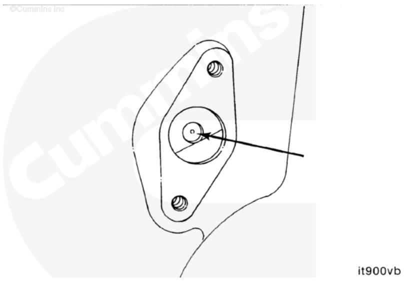


Loosen the intake valve adjusting screw and reset the valve to the proper clearance. Refer to Procedure 003-004 (/qs3/pubsys2/xml/en/procedures/41/41-003-004.html).

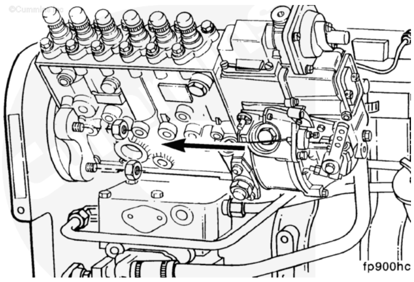
Rotate the crankshaft in the direction of the engine rotation until the pointer is aligned with the top dead center.



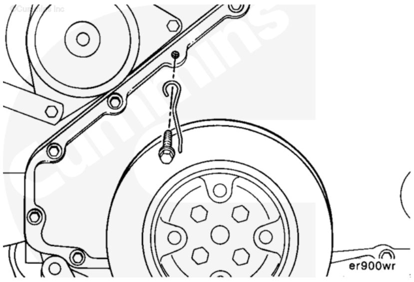
Look through the back side of the gear housing for the timing pin hole in the camshaft gear. If the hole is **not** visible, the crankshaft **must**, be rotated one revolution.



The engine is at the proper top dead center and the fuel pump can be installed for the correct timing. See the Install procedure.

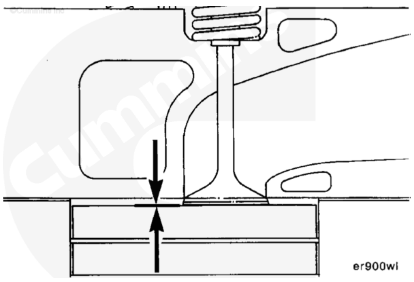


Remove the timing mark pointer from the front gear cover.



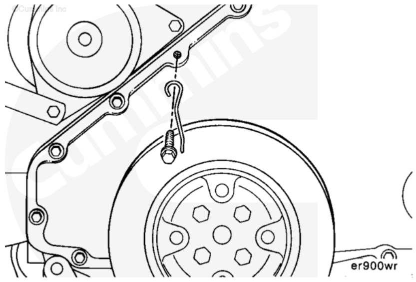
Automotive and Industrial

Use cylinder number 1 intake valve to make certain that the engine is at top dead center on the compression stroke for cylinder number 1. Refer to Procedure 001-049 (/qs3/pubsys2/xml/en/procedures/41/41-001-049.html).



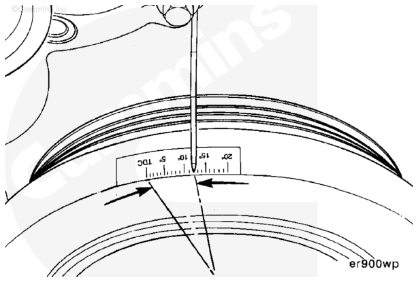
Fabricate a timing marker for the front of the engine.

Note : This can be done by forming a piece of 16 gauge wire that can be tightened under one of the gear cover capscrews. Sharpen the wire at the vibration damper end so that it comes to a point for better accuracy.



Attach a degree wheel or degree tape to the front of the vibration damper. Line the top dead center mark up with the pointer.

The degree wheel/tape **must** measure to an accuracy of at least ±1 degree.



⚠ WARNING ⚠

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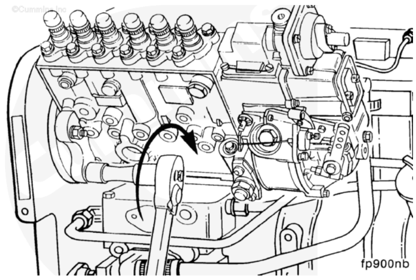
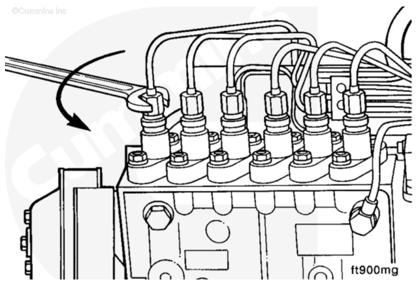


Install the fuel injection pump. See the Install section.

If the fuel injection pump is already installed, continued the procedure.

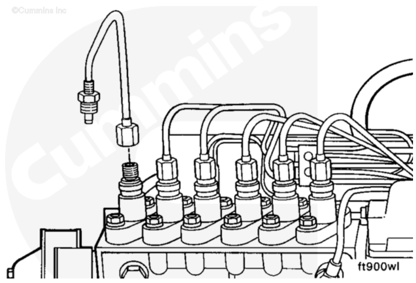
Remove the number 1 high-pressure fuel line from the fuel injection pump.

Note : Lines 2 through 6 must not be removed or loosened.



⚠ CAUTION ⚠

When attaching the fabricated tube, do not bend the number 1 high-pressure fuel line. This can cause the inside of the fuel line to flake and cause injector failure.



A short length of high-pressure line that is compatible with the fuel lines used on the engine **must** be bent in a “U” shape and installed onto the delivery valve holder of the fuel injection pump.

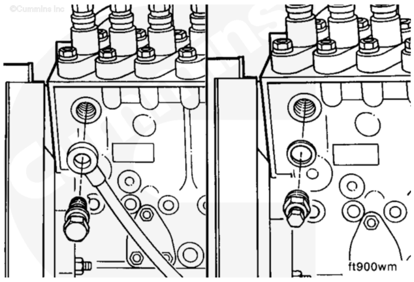
The line is used to observe when the fuel is or is **not** flowing through the delivery valve holder assembly.

Place a container under the tube to catch the fuel or drain the fuel back into the spill port pump.

Remove the overflow valve from the fuel injection pump.

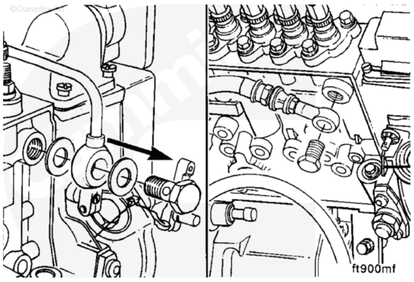
Install a 14-mm threaded plug and sealing washer into the fuel return port of the fuel injection pump.

Note : The fuel return port is located on the inboard front side of the fuel injection pump for automotive in-line applications and on the outboard front side for most industrial applications

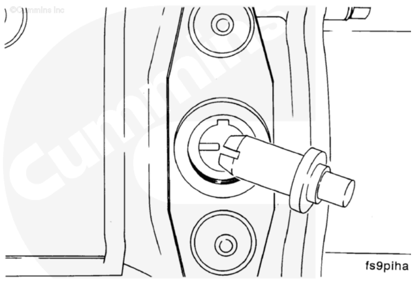


Remove the fuel supply line between the fuel filter head and the fuel injection pump.

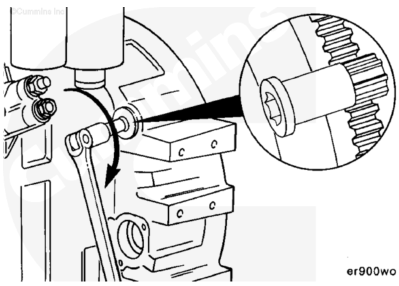
Attach the high-pressure outlet hose from the spill port pump cart to the fuel injection pump supply port.



Before continuing, make certain that the fuel injection pump timing pin is disengaged.



Rotate the crankshaft **counterclockwise**, as viewed from the front of the engine, to approximately 40 degrees before top dead center.



Adjust

Governor

⚠ WARNING ⚠

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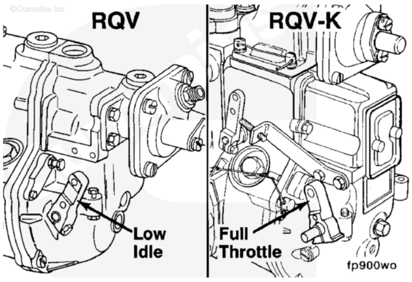


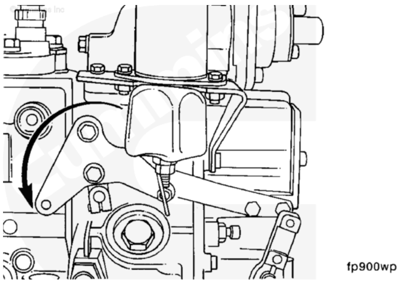
Note : The governor lever must be positioned before pressuring the fuel injection pump.

The RQV governor throttle lever **must** be in the low-idle lever position.

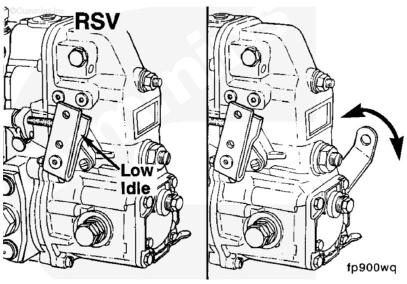
Automotive engines with a RQV-K governor throttle lever **must** be in the high-idle throttle position.

Industrial engines with a RQV-K governor throttle lever **must** be in the low-idle throttle position.

Both the RQV and RQV-K governor **must** have the shutdown lever in the full-run position.



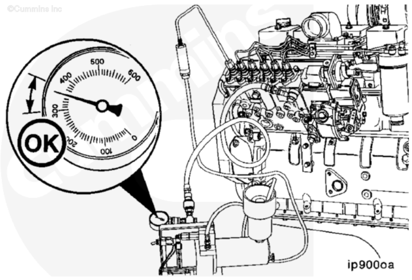
The RSV governor throttle lever **must** be in the low-idle position and the shutdown lever needs to be wired or locked in the 1/2-travel position.



Turn on the spill timing cart pump.

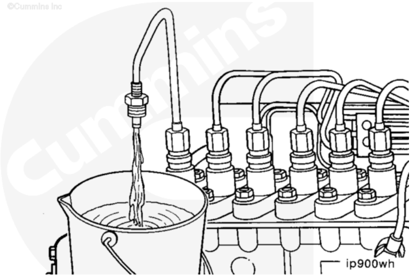
Check the fuel pressure.

Fuel Spill Timing Cart - Fuel Pressure		
kpa		psi
2068	MIN	300
2551	MAX	370



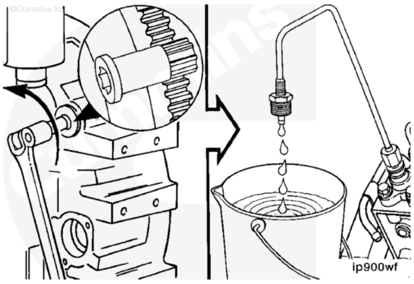
Note : The shutdown lever must be held in the required position before turning the spill cart pump on.

Fuel **must** be flowing out of the tube attached to the fuel injection pump. If the fuel is **not** flowing, check the procedures again carefully.



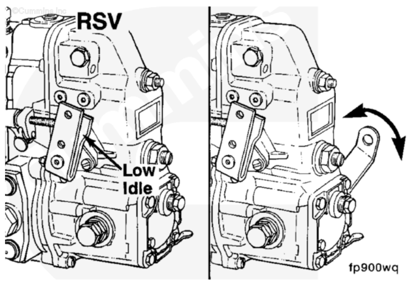
Slowly rotate the crankshaft **clockwise**, as viewed from the front of the engine, until fuel flow from cylinder number 1 begins.

The plunger number 1 element is now approaching port closure. Continue to rotate the crankshaft slowly until the flow is reduced to a drip. At the point that the steady stream of flow changes from a solid flow to a drip, stop. This is the static timing position of the fuel injection pump.

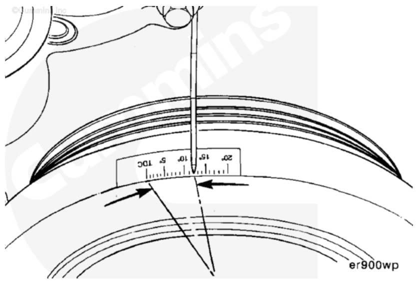


If the flow does **not** slow down to a drip:

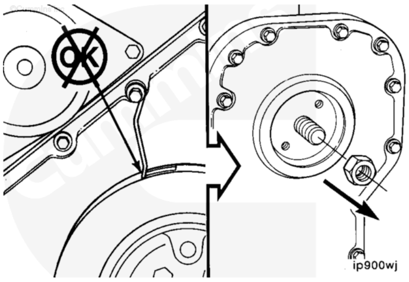
- Check the position of the governor lever.
- Make certain that the cylinder number 1 is before top dead center on the compression stroke.
- Turn off the spill port pump.



Check the degree wheel on the vibration damper to see what engine degree the timing pointer is indicating. This is spill port static timing. Compare this number to the timing specification for your particular application.



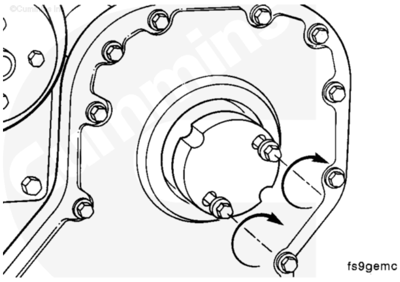
If the fuel injection pump static timing, as measured by the above method, is **not** within specification, remove the large nut that fastens the fuel injecting pump camshaft to the fuel pump drive gear.



If the crankshaft has rotated, turn on the spill port pump and rotate the crankshaft to find port closure.

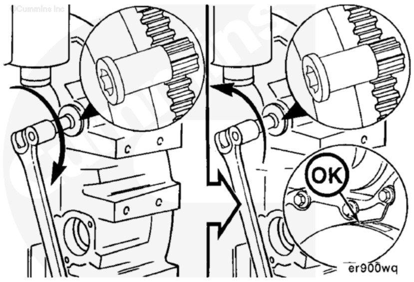
Turn off the spill port pump.

Use fuel pump gear puller, Part Number 3163381, with M8-1.25 x 50 capscrews, grade 8.8 or equivalent. Pull the fuel injection pump drive gear loose from the shaft.



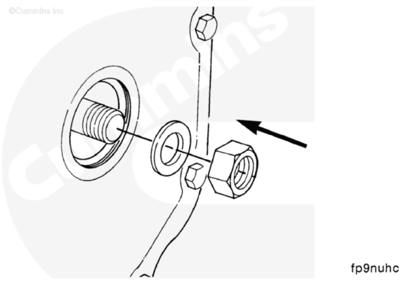
Slowly rotate the crankshaft **counterclockwise** about 40 degrees past the desired static timing specification.

Slowly rotate the crankshaft **clockwise** until the timing pointer indicates the desired static timing.



⚠ CAUTION ⚠

To prevent damage to the timing pins, do not exceed the torque value given. This is not the final torque value for the retaining nut.



Install and tighten the retaining nut and washer.

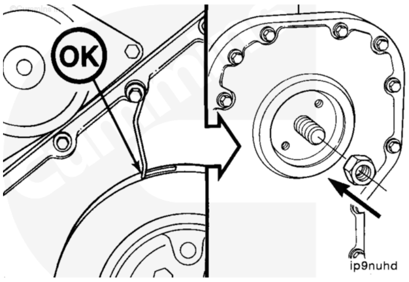
Measurements		
	n.m	in-lb
Torque Value	12	106

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

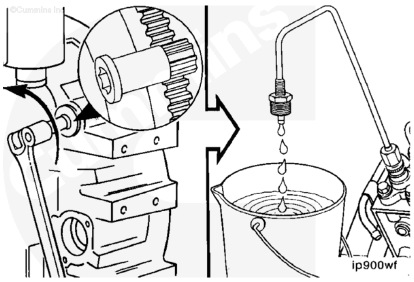
Failure to clean and dry the shaft and gear tapers thoroughly can result in timing shift to the retarded side after the engine is started and running under a load. This will result in low power, smoke, rough running, and engine damage.

Tighten the fuel injection pump drive nut.

Make certain that the static timing has **not** changed after the fuel injection drive nut is tightened to the required specification.

Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with QD Contact Cleaner, Part Number 3824510, by spraying into the gap between the shaft and the gear. Dry the surface with compressed air.

Tighten the fuel injection pump drive gear nut.



Torque Values:

Torque Value:
Nippondenso

1. 123 n•m [91 ft-lb]

Torque Value:
Bosch A Pump

1. 85 n•m [63 ft-lb]

Torque Value:
Bosch MW Pump

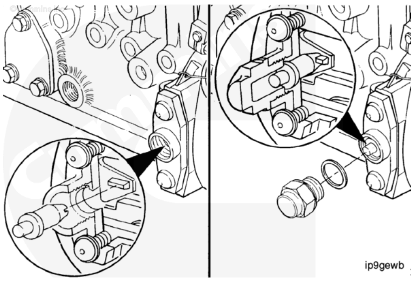
1. 105 n•m [77 ft-lb]

Torque Value:
Bosch P3000/P7100

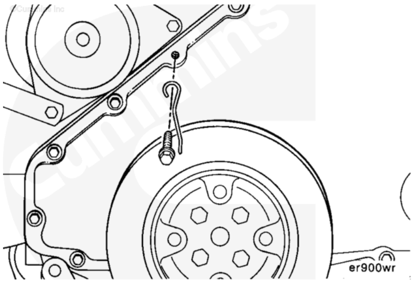
1. 195 n•m [144 ft-lb]

Repeat this procedure as needed until the timing matches the specification.

The fuel injection pump timing pin **must** fit over the injection pump pointer when the engine is at top dead center or on the compression stroke for the cylinder number 1. If it does **not**, the fuel injection pump **must** be adjusted by an authorized fuel injection pump shop or the fuel injection pump was installed incorrectly.



Remove the degree wheel and timing mark pointer.



Remove

Automotive and Industrial

⚠ WARNING ⚠

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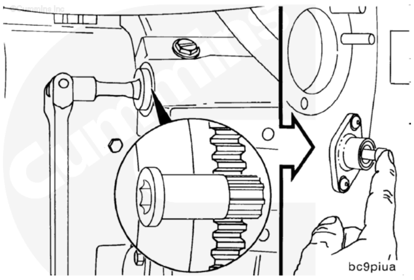
⚠ CAUTION ⚠

Do not use the timing pin to locate engine top dead center for marine applications. This can result in incorrect timing, poor engine performance, and engine smoke problems.

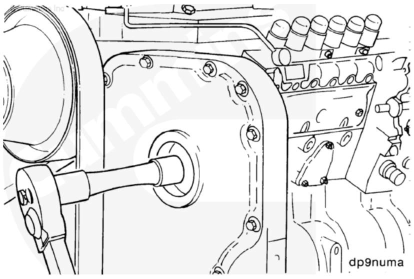


Locate top dead center for cylinder Number 1. Push the timing pin into the hole in the camshaft gear while slowly rotating the crankshaft with the barring tool, Part Number 3377371.

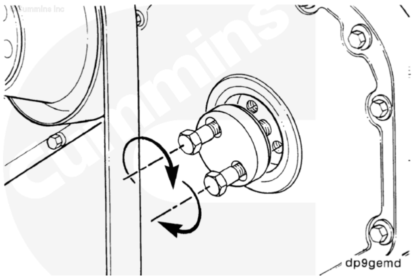
Note : Make certain that the timing pin is disengaged after locating top dead center.

Remove the front gear cover access cap.

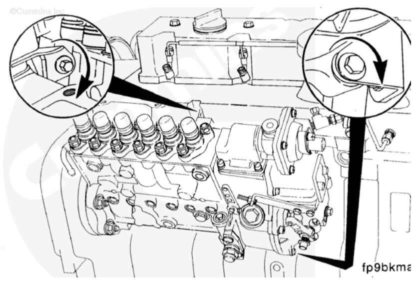
Remove the nut and washer from the fuel injection pump shaft.



Use fuel pump gear puller, Part Number 3163381, with M8-1.25 x 50 capscrews, grade 8.8 or equivalent. Pull the fuel injection pump drive gear loose from the shaft.

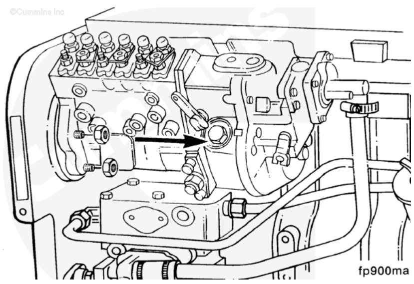


Engines equipped with the P7100, P3000, A, and MW fuel injection pump are equipped with support brackets that **must** be removed.



Remove the four fuel injection pump mounting nuts.

Remove the fuel injection pump.



Marine Applications

⚠ WARNING ⚠

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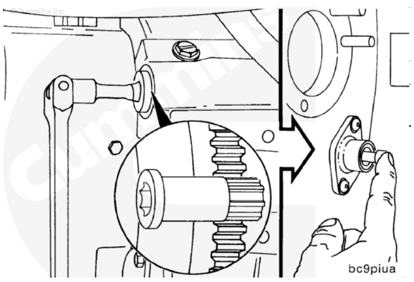


⚠ CAUTION ⚠

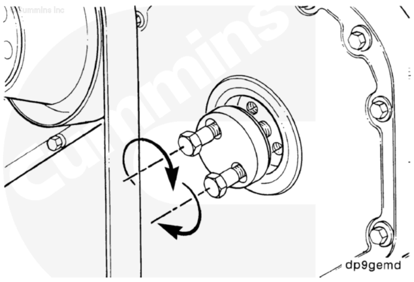
Do not use the timing pin to locate engine top dead center for marine applications. This can result in incorrect timing, poor engine performance, and engine smoke problems.

Locate top dead center for cylinder number 1. See the Time, Marine Applications section.

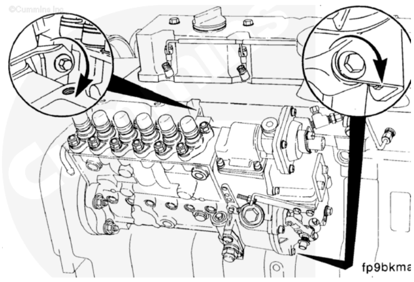
Remove the front gear cover access cap.

Remove the nut and washer from the fuel injection pump shaft.

Use fuel pump gear puller, Part Number 3163381, with M8-1.25 x 50 capscrews, grade 8.8 or equivalent. Pull the fuel injection pump drive gear loose from the shaft.

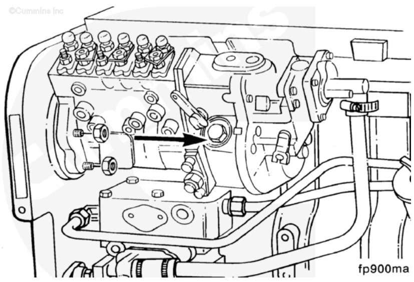


Engines equipped with P7100, P3000, A, or MW fuel injection pumps are equipped with support brackets that **must** be removed.



Remove the four fuel injection pump mounting nuts.

Remove the fuel injection pump.



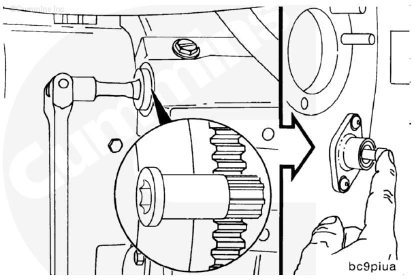
Install

Automotive and Industrial

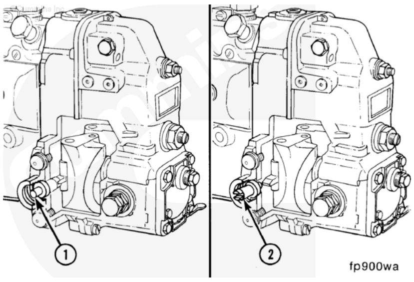
⚠ CAUTION ⚠

Do not use the timing pin to locate engine top dead center for marine applications. This can result in incorrect timing, poor engine performance, and engine smoke problems.

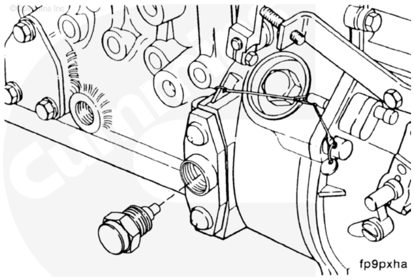
Make certain that the cylinder Number 1 is at top dead center.



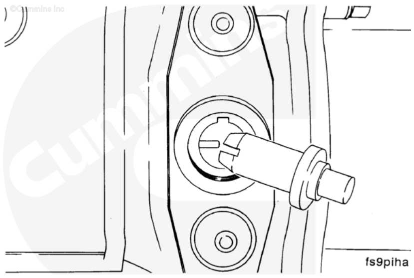
The fuel injection pump also has a timing pin (1), located in the governor housing, to position the fuel injection pump shaft to correspond with top dead center for cylinder Number 1. The timing pin **must** be reversed and stored in the housing (2) after the fuel injection pump is installed.



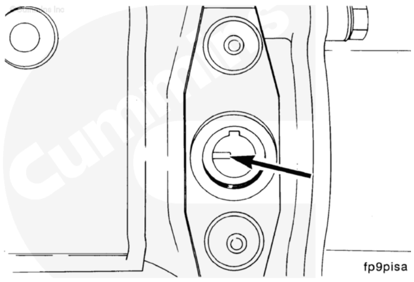
Remove the fuel injection pump timing pin access plug.



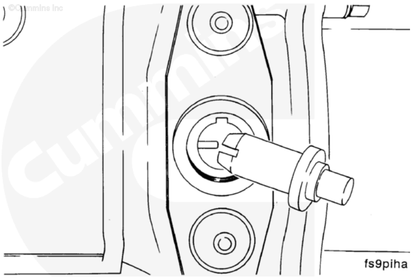
Remove the timing pin.



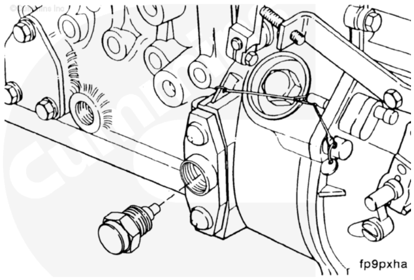
If the timing tooth is **not** aligned with the timing pin hole, rotate the fuel injection pump shaft until the timing tooth aligns.



Reverse the position of the timing pin so that the slot of the timing pin will fit over the timing tooth in the fuel injection pump.

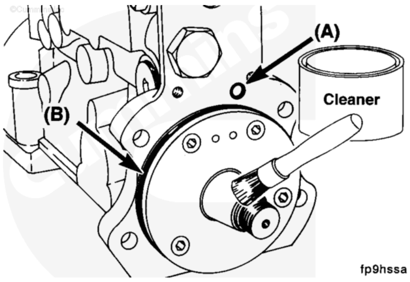


Install and secure the timing pin with the access plug.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



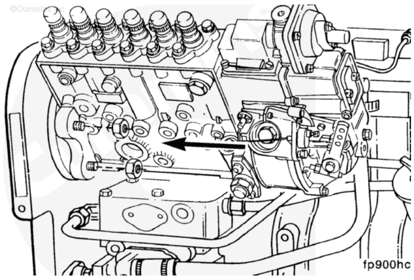
Make certain that the o-ring seals for the fill orifice and pilot are correctly installed and are **not** damaged.

Lubricate the mounting flange with clean lubricating engine oil.

Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with QD Contact Cleaner, Part Number 3824510, by spraying into the gap between the shaft and the gear.

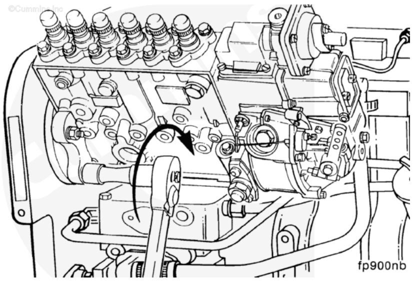
Dry the surface with compressed air.

Slide the fuel injection pump shaft through the drive gear, and position the fuel injection pump flange onto the mounting studs.



⚠ CAUTION ⚠

Do not pull the injection pump into the gear housing with the mounting nuts. Damage to the gear housing and fuel pump can result.



Install the fuel injection pump mounting nuts.

Install the support bracket, if equipped.

Torque Value:
Mounting nuts

1. 44 n•m [32 ft-lb]

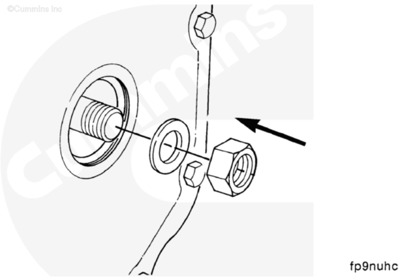
Torque Value:
Support bracket nuts

1. 32 n•m [24 ft-lb]

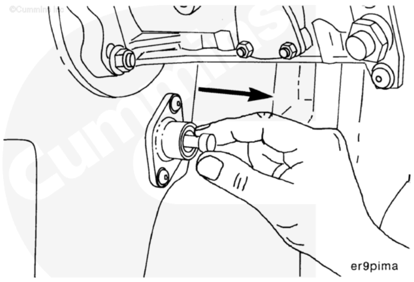
Note : To prevent damage to the timing pins, do not exceed the torque value given. This is not the final torque value for the retaining nut.

Install and tighten the fuel injection pump retaining nut and washer.

Torque Value: 12 n•m [106 in-lb]

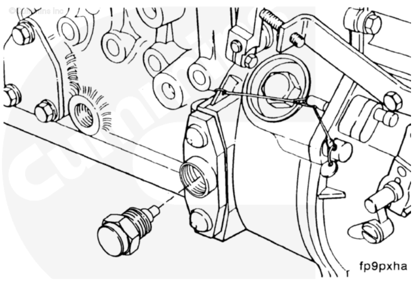


Disengage the engine timing pin.



⚠ CAUTION ⚠

The governor housing must be pre-lubricated before engine operation. Failure to do so will result in premature governor wear.



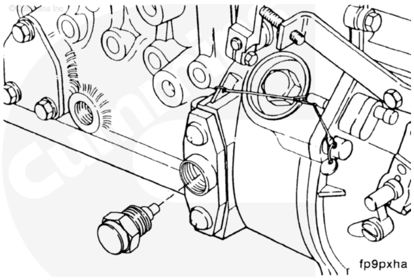
Remove the access plug.

Add the following quantity of clean engine oil:

- RSV 450 ml [0.48 qt]
- RQV 750 ml [0.79 qt]
- RQVK 750 ml [0.79 qt].

Install the access plug.

Torque Value: 28 n•m [21 ft-lb]



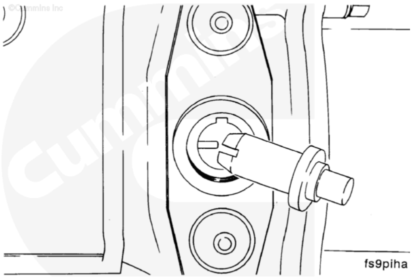
Remove the fuel injection pump timing pin plug.

Reverse the position of the timing pin.

Install the timing pin, plug, and sealing washer.

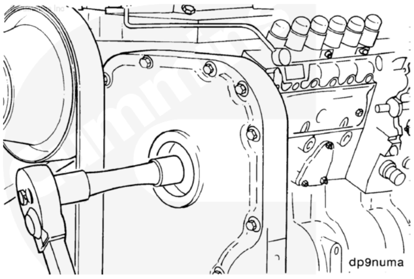
Tighten the timing pin plug.

Torque Value: 15 n•m [133 in-lb]



⚠ CAUTION ⚠

Failure to clean and dry the shaft and gear tapers thoroughly can result in timing shift to the retarded side after the engine is started and running under a load. This will result in low power, smoke, rough running, and engine damage.



Tighten the fuel injection pump drive gear nut.

Torque Value:
Nippondenso

1. 123 n•m [91 ft-lb]

Torque Value:
Bosch® A pump

1. 85 n•m [63 ft-lb]

Torque Value:
Bosch® MW pump

1. 105 n•m [77 ft-lb]

Torque Value:
Bosch® P3000/P7100

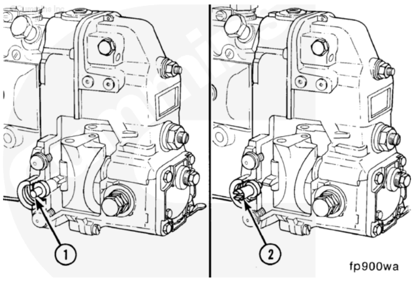
1. 195 n•m [144 ft-lb]

Install the gear cover access cap hand-tight.

Marine Applications

⚠ CAUTION ⚠

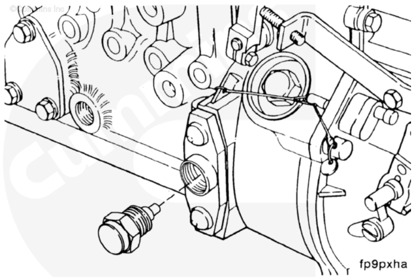
Do not use the timing pin to locate engine top dead center for marine applications. This can result in incorrect timing, poor engine performance, and engine smoke problems.



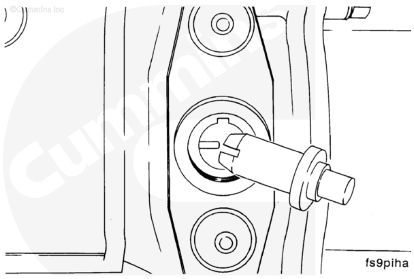
Make certain that the cylinder number 1 is at top dead center. See the Time, Marine Application procedure.

The fuel injection pump also has a timing pin (1), located in the governor housing, to position the fuel injection pump shaft to correspond with top dead center for cylinder number 1. The timing pin **must** be reversed and stored in the housing (2) after the fuel injection pump is installed.

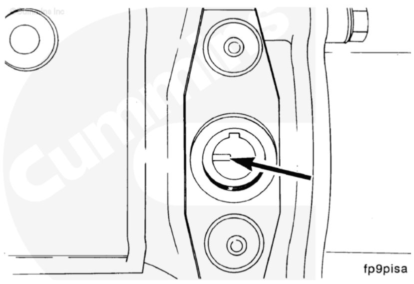
Remove the fuel injection pump timing pin access plug.



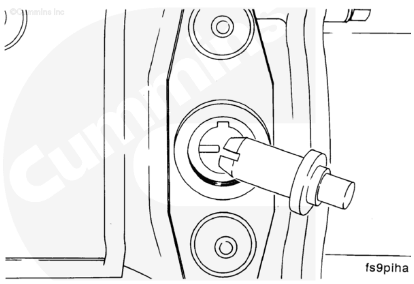
Remove the timing pin.



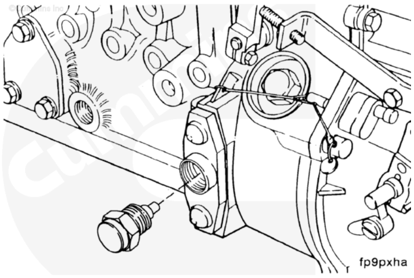
If the timing tooth is **not** aligned with the timing pin hole, rotate the fuel injection pump shaft until the timing tooth aligns.



Reverse the position of the timing pin so that the slot of the timing pin will fit over the timing tooth in the fuel injection pump.

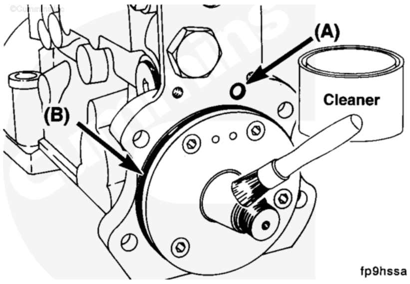


Install and secure the timing pin with the access plug.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

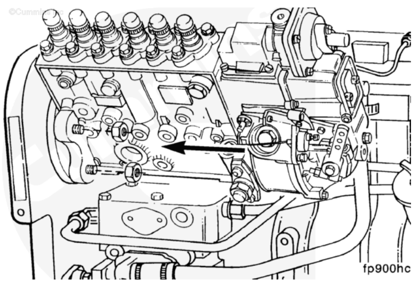
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Make certain that the o-ring seals for the fill orifice and pilot are correctly installed and are **not** damaged.

Lubricate the mounting flange with clean lubricating engine oil.

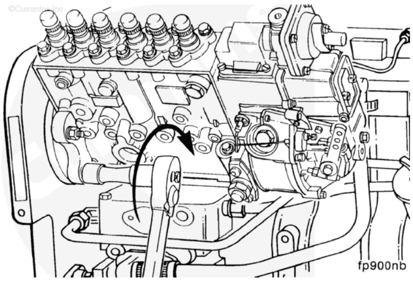
Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with QD Contact Cleaner, Part Number 3824510, by spraying into the gap between the shaft and gear. Dry the surface with compressed air.

Slide the fuel injection pump shaft through the drive gear, and position the fuel injection pump flange onto the mounting studs.



⚠ CAUTION ⚠

Do not pull the injection pump into the gear housing with the mounting nuts. Damage to the gear housing and fuel pump can result.


fp900nb

Install the fuel injection pump mounting nuts.

Install the support bracket, if equipped.

Torque Value:
Mounting nuts

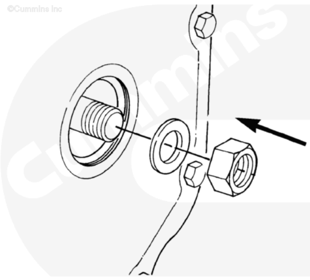
1. 44 n•m [32 ft-lb]

Torque Value:
Support bracket nuts

1. 32 n•m [24 ft-lb]

⚠ CAUTION ⚠

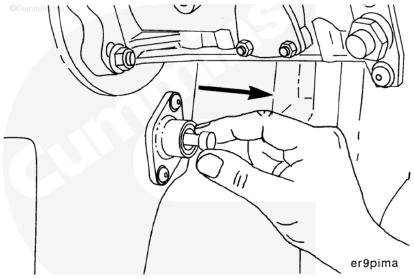
To prevent damage to the timing pins, do not exceed the torque value given. This is not the final torque value for the retaining nut.


fp9nuhc

Install and tighten the fuel injection pump retaining nut and washer.

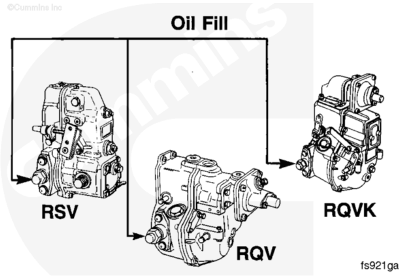
Torque Value: 12 n•m [106 in-lb]

Disengage the engine timing pin.


er9pima

⚠ CAUTION ⚠

The governor housing must be pre-lubricated before engine operation. Failure to do so will result in premature governor wear.



fs921ga

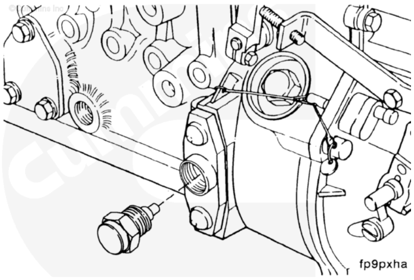
Remove the access plug.

Add the following quantity of clean engine oil.

- RSV 450 ml [0.48 qt]

Install the access plug.

Torque Value: 28 n•m [21 ft-lb]



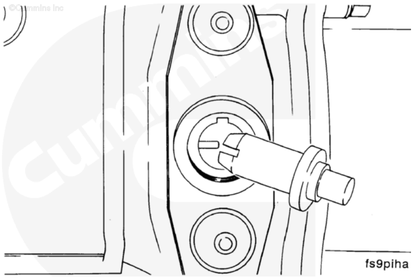
Remove the fuel injection pump timing pin plug.

Reverse the position of the timing pin.

Install the timing pin, plug, and sealing washer.

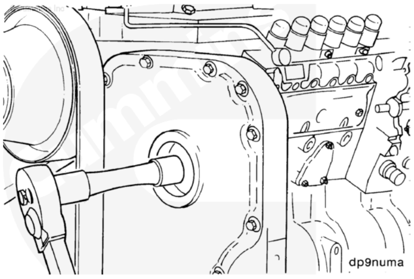
Tighten the timing pin plug.

Torque Value: 15 n•m [133 in-lb]



⚠ CAUTION ⚠

Failure to clean and dry the shaft and gear tapers thoroughly can result in timing shift to the retarded side after the engine is started and running under a load. This will result in low power, smoke, rough running, and engine damage.



Tighten the fuel injection pump drive gear nut.

Torque Value:
Nippondenso

1. 123 n•m [91 ft-lb]

Torque Value:
Bosch® P3000/P7100

1. 195 n•m [144 ft-lb]

Install the gear cover access cap hand-tight.

Finishing Steps

- Install the governor oil supply line.
- Install the air fuel control air line.
- Install the fuel shutoff solenoid. Refer to Procedure 005-043 (/qs3/pubsys2/xml/en/procedures/41/41-005-043.html).

- Install the control linkage. See the OEM service manual.
- Install the injector supply lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html).
- Install the fuel supply lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/41/41-006-024.html).

The MW-type fuel injection pump **not** equipped with engine-side fuel return option **must** be vented after installation. Loosen the vent screw located near the front on the side nearest the engine. Place the fuel control in the run position. Crank the engine so that air can bleed from the fuel injection pump. Then, tighten the vent screw.

Earlier MW-type fuel injection pumps were **not** equipped with a vent screw. Remove the large plug from the location described above to vent the fuel injection pump.

A-type fuel injection pumps are self-venting. All P-type pumps were assembled with the engine-side vent option.

Last Modified: 12-Aug-2019